



Document Management System



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Title: Intubation & Post Intubation Care

Introduction:

Endotracheal intubation is a common procedure used during resuscitation and stabilization of critically ill children. The most common reason for invasive ventilation among pediatric ICU patients is respiratory distress.

Indications for invasive ventilation

Oxygenation abnormalities

- Cyanosis with $FiO_2 > 0.6$
- $PaO_2 < 52$ mmHg with $Fi O_2 > 0.6$
- Need for PEEP/Failure to non-invasively oxygenate
- Excessive work of breathing

Ventilation abnormalities

- Apnoea
- $GCS < 8$ to protect the airway
- Increased airway resistance/obstruction with rising $PaCO_2$ and acidosis
- Decreased ventilatory drive with $PaCO_2 > 55$ or rising $PaCO_2$
- Neuromuscular diseases e.g. Guillan-Barré with $FEV < 15$ ml/kg
- Respiratory muscle dysfunction with Respiratory muscle fatigue and Chest wall abnormalities

Other indications

- Permit sedation or neuromuscular blockade
- Decrease systemic + myocardial O_2 consumption
- Reduce intracranial pressure
- Post arrest

Defined as unwanted adverse events that occur as a consequence of the tracheal intubation encounter or desaturation < 80%.

Incidence of TIAE is 20% and 6% of these adverse events are severe .Most common TIAE is esophageal intubation. Severe TIAE include severe desaturation, hypotension needing intervention, laryngospasm, air leak and cardiac arrest.

Incidence of TIAE increases adverse PICU outcomes including mortality and increase in PICU stay.

Risk factors for TIAE

The following are known risk factors for TIAE:

- Patient 's factors including infants, diagnosis with respiratory disease, severity of illness and hemodynamic instability(physiological difficult airway). Physiological difficult airway is when the glottis view is normal in the presence of physiological abnormalities such as sepsis and hypoxia which leads to increase adverse events.
- History of difficult airway
- Experience of the intubator .Less experienced intubator is associated with increase in TIAE
- Lack of safety equipment (capno-graphy and alternative airway)
- Lack of planning of the process

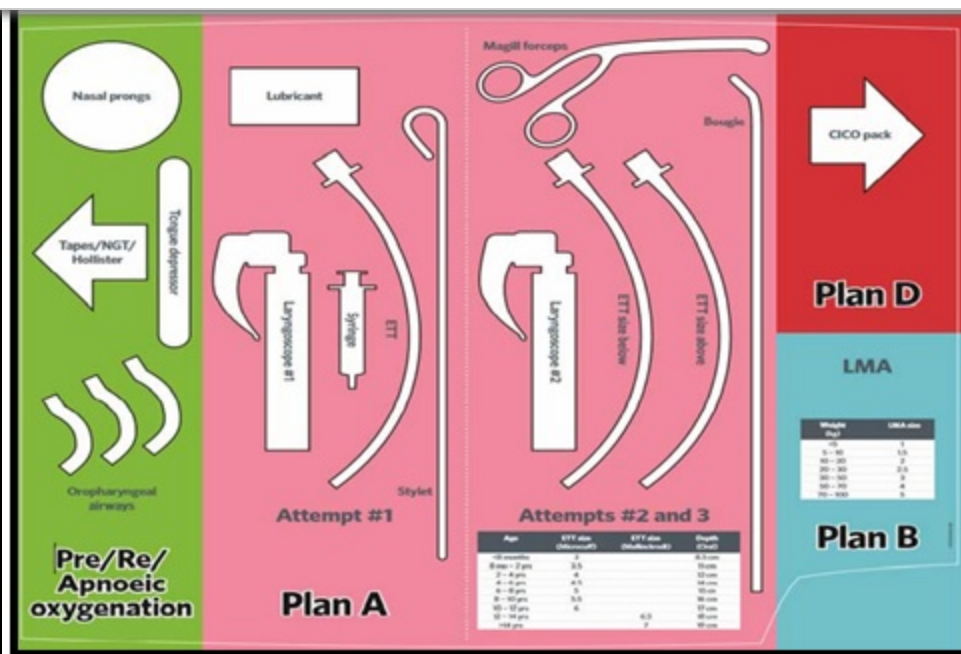
Planning and preparation for intubation

In order to reduce TIAE, the medical team need to:

- 1.Optimise patient's condition by improving related hemodynamic instability and use of pre-oxygenation
- 2.Use the most experienced intubator for higher risk intubation such as in children with cardiogenic and septic shock
- 3.Plan the process of intubation

Before intubation, prepare medical team, patient, IV drugs, monitors and equipment. See the following chart.

Emergency tracheal Intubation Checklist (For team leader use prior to every Emergency Intubation) If any difficulty anticipated CALL FOR HELP			
TEAM	PATIENT	IV DRUGS MONITORS	EQUIPMENT
1.Notify Consultant/Senior Registrar 2.Verbalize indication for intubation 3.Allocate roles 4.Confirm intubation plan A. Initial tracheal intubation attempts X 3 B. Final tracheal intubation attempt C. Rescue plan to maintain oxygenation D. Rescue plan to surgical airway 5. Assign lead for the post intubation-debrief	1.Optimize hemodynamics. Consider: Fluid bolus Inotropes/ Vasopressors Bolus dose vasopressors drawn up 2.Optimize pre-oxygenation Consider: 100% FIO2 PEEP via t-piece Apneic Oxygenation (NC):2l/kg/ min (Max 15l/min) Elevate head of bed 3.Optimize position, consider: <1 year: towel under shoulders >8 years towel/pillow under head	1.IV access functioning 2.Intubation drugs/dose chosen and drawn up 3.Cardiac monitoring 4.BP (2minute cycle) 5.SpO2 (set audible) 6.ETCO2 7. Post intubation sedation drawn up	1.T-piece/face mask 2.Suction functioning (Yankauer and flexible) 3.Airway equipment template complete 4.Video-Laryngoscope at bedside/ turned on



Calculation of the appropriate size ET tubes:

- Uncuffed endotracheal tube (mm ID) = (age in years / 4 + 4)
- Cuffed endotracheal tube (mm ID) = age / 4 + 3

Depth of insertion of ET at teeth

(Age / 2) + 12 at teeth or internal diameter [ID] of the tube times 3

Pre-oxygenation:

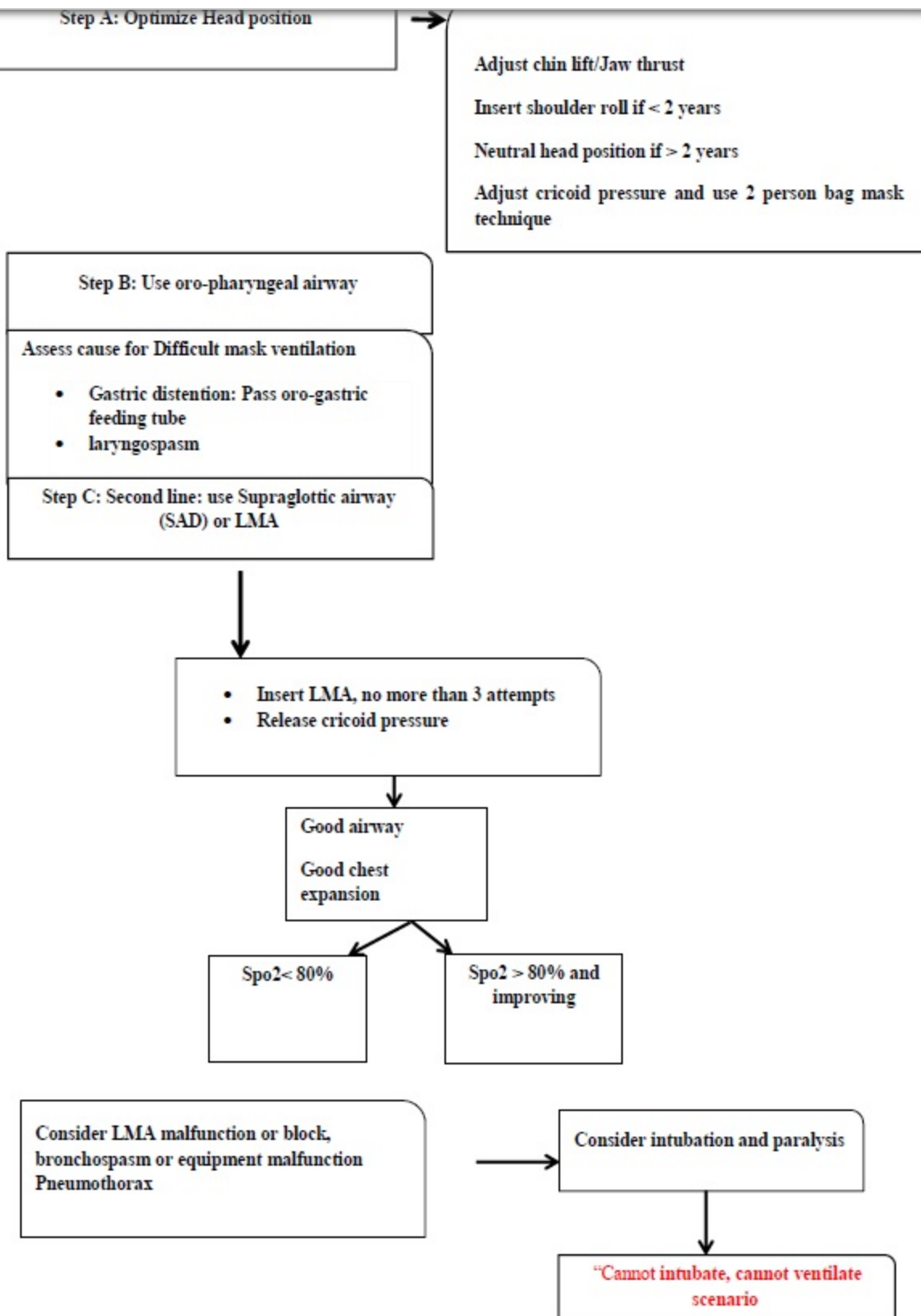
Pre-oxygenate using nasal cannula, High flow nasal cannula or using NIV if child is already on NIV for at least 2 minutes to achieve oxygen saturation of 100%. Pre-oxygenation leads to reduced desaturation during intubation. This is important in small infants and sick children admitted with sepsis or hemodynamic instability to achieve safe apnea times during intubation.

Risk factors for difficult airway

- Anatomical abnormalities including mandibular hypoplasia, micrognathia, maxillary/midfacial hypoplasia, cleft palate
- Acutely altered airway: such as in epiglottitis, croup, angioedema, neck masses and anterior mediastinal masses
- Children with underlying congenital syndromes including Down syndrome, Pierre Robin, MPS (mucopolysaccharidosis) and Treacher Collins syndrome
- Previous history of difficult airway
- Physiological difficult airways: difficulties due to associated cardiovascular compromise due to sepsis or underlying cardiac disease or metabolic acidosis and shock where use of sedative agents for intubation may lead to hemodynamic compromise and arrest.

Pathway for management of children with difficult airways

Pathway for difficult bag mask ventilation



Pathway for cannot intubate, cannot ventilate:

Step A Continue to attempt oxygenation and ventilation

- FiO_2 1.0
- Optimise head position and chin lift/jaw thrust
- Insert oropharyngeal airway or SAD (e.g. LMA™)
- Ventilate using two person bag mask technique
- Manage gastric distension with an OG/NG tube

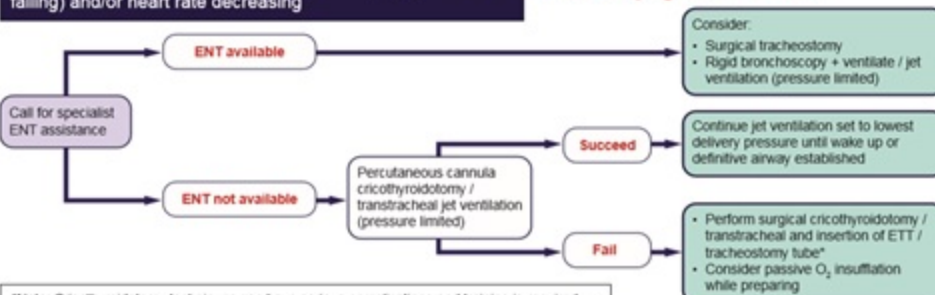
Step B Attempt wake up if maintaining $\text{SpO}_2 > 80\%$

If rocuronium or vecuronium used, consider sugammadex (16mg/kg) for full reversal

Prepare for rescue techniques in case child deteriorates

Step C Airway rescue techniques for CVCV ($\text{SpO}_2 < 80\%$ and falling) and/or heart rate decreasing

Call for help again if not arrived



Cannula cricothyroidotomy

- Extend the neck (shoulder roll)
- Stabilise larynx with non-dominant hand
- Access the cricothyroidotomy membrane with a dedicated 14/16 gauge cannula
- Aim in a caudal direction
- Confirm position by air aspiration using a syringe with saline
- Connect to either:
 - adjustable pressure limiting device, set to lowest delivery pressure
- or
- 4Bar O_2 source with a flowmeter (match flow l/min to child's age) and Y connector
- Cautiously increase inflation pressure/flow rate to achieve adequate chest expansion. Wait for full expiration before next inflation
- Maintain upper airway patency to aid expiration

SAD = supraglottic airway device

Post intubation care

- Check for ET Co_2 trace (exclude esophageal intubation)
- Listen over the chest for symmetrical air entry and
- Obtain CXR

Position of ET on CXR

Optimum position for ET 1.5 cms above carina

Or at level of T2-T4

Ensure that ET secure

- Record the ET distance at lips
- Make sure that strapping of ET is secure, if becomes loose it needs to be re-strapped
- 2 nurses are needed to do the strapping
- Extra dose of sedation or paralysis may be needed during tube strapping

Ensuring adequacy of ventilation

- Observe for symmetrical chest expansion
- Ventilated patients need to have end tidal co_2 monitoring
 - Nurses to listen over the chest every 2 -4 hours and to inform the resident if any changes

Common drugs used for intubation

Common premedications and induction agents

Pentanyl	1-2 mg/kg IV	newborns
Morphine	0.05mg-0.1 mg/kg	
Midazolam	0.1 mg/kg	Use with caution in hypotensive patients
Thiopental	3-5 mg/kg IV	Lower dose to 1 mg/kg in children with hypotension or avoid it , useful in children with raised ICP
Ketamine	1-2 mg/kg IV, 4 mg/kg IM	
Propofol	1-2 mg/kg IV	Avoid in hypotensive patients
Paralytics		
Succinylcholine	2 mg/kg IV	Have atropine drawn up and ready
Pan/vecuronium	0.2 mg/kg IV	May increase to 0.3 mg/kg of vecuronium for RSI. (0.1 mg/kg for maintenance of paralysis)
Rocuronium	1.0 mg/kg IV	(for rapid sequence intubation (RSI))

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